

Improve Performance — Cloud & Client

SUMMARY

The phone is a signal processor, not just a camera. Performance improvements come from two places: better signal quality captured on-device, and faster inference in the cloud. Together they reduce total assessment time and failed assessments.

TECH STACK

Android - MediaPipe · Vulkan · Kotlin
iOS - Vision framework · Metal · Swift/SwiftUI

APPROACH & RECOMMENDATION

Client: Metal (iOS) and Vulkan (Android) GPU frameworks run the signal extraction pipeline on-device. MediaPipe tracks face landmarks continuously. ROI extraction targets forehead and cheeks. POS + CHROM fusion extracts the blood volume pulse signal. A signal quality gate rejects low-quality frames before transmission, only clean signal reaches the cloud.

Cloud: warm pool means inference begins the moment signal arrives, no cold start wait. Cleaner incoming signal means the model runs faster with higher confidence.

The ISP (image signal processor) has auto-exposure, auto-white-balance, H.264 compression which degrades the rPPG signal. The on-device pipeline is designed to work around this.

ALTERNATIVES CONSIDERED & REJECTED

All compute in cloud, raw video transmitted, simpler day one but slower, phase 1 reality while stateless refactor is in progress. Shared C++ core across iOS and Android, reduces duplication but sacrifices platform-native performance.

Release Safely at Pace

SUMMARY

Release controls that don't slow the process. Everyone knows what is happening and consistency drives the right behaviour. In a regulated medical device the control is not optional, it is the evidence regulators need to trust what you ship.

TECH STACK

GitHub Actions · ACR · TestRail · SonarCloud
· Snyk · Syft · Qualio

APPROACH & RECOMMENDATION

Pipeline gates in order: build and unit tests, SonarCloud static analysis, Snyk dependency scan, AI code review, Syft SBOM generation, TestRail trigger, human approval to production. No automated promotion to production.

Three-layer test strategy: unit and integration tests automated via GitHub Actions. System verification in TestRail, requirement to test case to execution to result, this is the IEC 62304 regulatory artefact. Clinical performance validation to FDA submission standard, separate from software verification.

Every sprint produces a release candidate. Production requires PO sign-off, QA gate, and regulatory checklist.

ALTERNATIVES CONSIDERED & REJECTED

Informal test evidence back-filled before submission. The cost of back-filling is always higher than the cost of discipline. Rejected. Fully automated promotion to production. Not compatible with regulated software sign-off requirements. Rejected. Compliance tooling separate from the delivery pipeline. Creates two systems of record, engineers treat compliance as something that happens to them rather than something they own. Rejected.

Trade-off: compliance gates add friction. The compliance agent makes traceability linkage visible at PR time — engineers see the gap before it becomes a submission problem, not after. AI code review without measuring action rate risks becoming noise engineers route around.